A **thick-client** is one which:

- chooses an application provided by the server
- possibly carries out some processing before running the application on the server and also after receiving output from the application
- alternatively, possibly downloads the application from the server and runs the application itself.



TIP

In thick-client mode the processing on the client can be controlled by the use of a scripting language. You do not need to know any details of this.



KEY TERM

Thick-client: a client that carries out at least some of the processing itself

The client-server approach is the choice in the following circumstances.

- The server stores a database which is accessed from the client system.
- The server stores a web application which allows the client system to find or, sometimes, supply information.
- The server stores a web application which allows the client system to carry out an e-commerce or financial transaction.

File sharing

If a user uploads files to a file server then the client-server operation can be used by another user to download these from the server.

An alternative mode of operation for sharing files is peer-to-peer networking. Instead of having one server that many clients access, a peer-to-peer network operates with each peer (networked computer) storing some of the files. Each peer can therefore act as a client and request a file from another peer or it can act as a server when another peer requests the download of a file.

The peer-to-peer model has several advantages compared to client-server file downloading:

- it avoids the possibility of congestion on the network when many clients are simultaneously attempting to download files
- parts of a file can be downloaded separately
- the parts are available from more than one host.

The client-server model has the following advantages.

- It allows an organisation to control the downloading and use of files.
- The files can be better protected from malware attacks because the files are stored on one server which will be regularly scanned using appropriate anti-virus software.

2.02 Network topologies

There are five requirements for a data communications system: a sender, a receiver, a transmission medium, a message and a protocol (see Chapter 17 for details about protocols). A transmission medium can be air (e.g. for WiFi) or cables (e.g. for Ethernet). Data can be sent through the medium in different modes:

- simplex mode where data flow is one-way only
- half duplex where data can flow either way but not simultaneously
- full duplex where simultaneous both-ways data flow is possible.

A 'message' is any type of data, which can be sent as either:

- a broadcast, which is a one-to-all communication (as used traditionally for radio and television)
- a multicast, which is from one source to many destinations
- a unicast, which is a one-to-one communication.

A data communications system may consist of a single isolated network. There are several possibilities for the **topology** of an isolated network. The simplest of these is where two systems are connected by a network link as shown in Figure 2.01. This is an example of a point-to-point connection, which is a dedicated link. Transmission might be simplex or duplex and a message can only be unicast.



Figure 2.01 A point-to-point network

Early LAN topologies used either a ring or a **bus topology**. We don't need to cover the ring topology as it is not used very often now. A bus topology has only one link but it is shared by a number of end-systems and is therefore described as a multi-point connection. The configuration is shown in Figure 2.02. There is no direct connection between any pair of end-systems. A message must therefore be broadcast even though it might only be intended for one **end-system**. The topology is resilient because a fault in an end-system or in the link to it does not affect the use of the network by the other end-systems.



Figure 2.02 A bus network

An example of a fully-connected **mesh topology** is shown in Figure 2.03. In this configuration, each end-system has a point-to-point connection to each of the other end-systems. Transmission is duplex; messages might be unicast, multicast or broadcast.

KEY TERMS

Topology: the configuration of a network that defines how the various devices on the network are connected

Bus topology: contains one shared link to which all devices are attached

End-system: a computer or server connected to a network

Mesh topology: contains direct links between devices

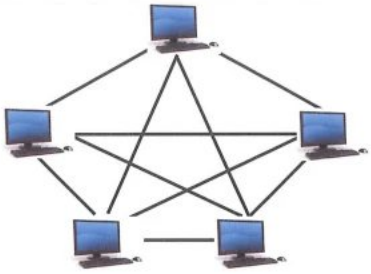


Figure 2.03 A mesh network.

Figure 2.03 shows end-systems connected in a mesh topology but this is unrealistic because of the amount of cabling required. A mesh topology can be used when individual LAN switches are connected in a network. The topology is essential for the connection of routers within the infrastructure of the Internet.

The final possibility is a **star topology** which is shown in Figure 2.04.

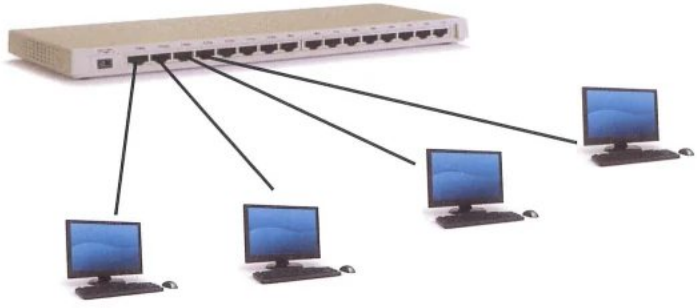


Figure 2.04 A star topology

KEY TERM

Star topology: each end-system is linked to a central device

Figure 2.04 could have been drawn so that it looked like a star but has been drawn to show the physical configuration that is used in a real life installation. In a star topology, each end-system has a point-to-point connection to the central device. Transmission is duplex and messages from the central device might be unicast, multicast or broadcast. As with the bus topology, the failure of an end-system, or its link, leaves the other end-systems unaffected. However, the central device must not fail.

In the bus topology most of the end-systems might be user workstations and the others are servers. However, in the star topology, the end-systems might be user workstations or servers but the central device is different. It is a specialised device with the purpose of connecting other devices in the network. Currently, the star topology is the usual way to configure a network. There are several reasons for this. The most important is that the central device can be used to connect the network to other networks and, in particular, to the Internet.

Discussion Point:

Which network topologies have you used? You might wish to defer this discussion until you have read about network devices later in this chapter.

In a situation where several LANs are connected, they can have different topologies or supporting technologies. This collection of LANs then becomes a **hybrid network**. A special connecting device is needed to ensure that the hybrid network is fully functional. It is often an advantage to be able to connect a new topology LAN to existing LANs where it is not sensible or not possible to use the existing topology for the new LAN. An example is when a wired LAN is already installed but a new wireless LAN is to be connected to it.

KEY TERMS

Hybrid network: a collection of connected LANs where some of them have different topologies or supporting technologies

Cable: a transmission using copper wire or fibre-optic

Bandwidth: a measure of the amount of data that can be transmitted per second

2.03 Transmission media

Cable

A network **cable** can be twisted pair, coaxial or fibre-optic. The twisted pair and coaxial cables both use copper for the transmission medium. In discussing suitability for a given application there are a number of factors to consider. One of these is the cost of the cable and connecting devices. Another is the best **bandwidth** that can be achieved. The bandwidth governs the possible data transmission rate. There are then two factors that can cause poor performance: the likelihood of interference affecting transmitted signals and the extent of attenuation (deterioration of the signal) when high frequencies are transmitted. These factors will dictate whether repeaters or amplifiers are needed in transmission lines and how many will be needed. Table 2.01 shows some comparisons of the different cable types.

	Twisted pair	Coaxial	Fibre-optic
Cost	Lowest	Higher	Highest
Bandwidth or data rate	Lowest	Higher	Much higher
Attenuation at high frequency	Affected	Most affected	Least affected
Interference	Worst affected	Less affected	Least affected
Need for repeaters	More often	More often	Less often

Table 2.01 Comparisons between cable types

You need to understand that for each of the three types of cabling there are defined standards for different grades of cable which must be considered when you decide which type of cable to use. Fibre-optic cable performs best but costs more than the other kinds. For a new installation the improved performance of fibre-optic cable is likely to be the factor that governs your choice. However, where copper cable is already installed the cost of replacement by fibre-optic cable may not be justified.

Specifically, the web consists of an enormous collection of websites each having one or more web pages. The special feature of a web page is that it can contain hyperlinks which, when clicked, give direct and essentially immediate access to other web pages.

Cloud computing

Cloud computing is the provision of computing services usually via the Internet. An organisation may choose to establish its own **private cloud**. In this case there are three possible approaches:

- The organisation takes full responsibility for creating and managing the cloud installed on-site and connected to a private network
- The organisation outsources to a third-party the creation and management of an on-site installation connected to a private network
- The organisation outsources the creation and management of an Internet accessible system by a third-party.

The alternative is a **public cloud**. This is created, managed and owned by a third-party cloud service provider.

The services provided by a cloud are familiar ones provided by file servers and application servers. They are accessible via a browser and therefore accessible from any suitable device in any location. A public cloud can be accessed by an individual user or by an organisation. One major difference is the scale of the systems. The provision is established using large mainframe computers or server farms. The services provided can be characterised as being one of:

- infrastructure provision
- platform provision
- software provision

Many of the advantages to a cloud user arise from the fact that the cloud does not have the limitations that the systems already available have. For the infrastructure provision, the advantages include the better performance when running software and the increased storage capacity. For the platform provision, the cloud can offer facilities for software development and testing. For the software provision, the cloud will be able to run applications that require high performance systems. Alternatively, it could be that the costs to a company of buying and installing a software package themselves would be far too high. The other advantage is the familiar one with regard to outsourcing. The cloud user no longer needs technical expertise.

The disadvantages to a cloud user relate to the use of a public cloud. The cloud service provider has complete access to all of the data stored on the cloud. The cloud user cannot be sure that their data is not being shared with third-parties. This is a concern with regard to data privacy. The security of the data stored is an issue; the cloud service provider is being relied on to ensure data cannot be lost.



KEY TERMS

Private cloud: owned by and only accessed by an organisation

Public cloud: owned by a cloud service provider for general access